

Customer Retention & Churn Analysis

Business Intelligence Report — Telco Customer Churn Dataset

Prepared for	Internship Submission — Data Analytics
Dataset	IBM Telco Customer Churn (7,043 customers, 21 attributes)
Report Date	2026
Tools Used	Python (Pandas, Seaborn, Matplotlib, Scikit-learn), Power BI, Jupyter

Customer Retention & Churn Dashboard

IBM Telco Customer Churn - Interactive Business Performance Overview

7,043 CUSTOMERS

CONTRACT

All

INTERNET SERVICE

All

PAYMENT METHOD

All

SENIOR CITIZEN

All

TENURE GROUP

All

Reset Filters

TOTAL CUSTOMERS

7,043

CHURN RATE

26.54%

RETENTION RATE

73.46%

AVG MONTHLY CHARGES

\$64.76

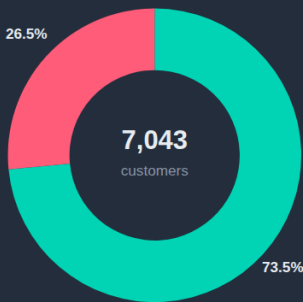
MONTHLY REVENUE LOST

\$139,131

AVG TENURE

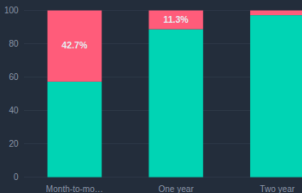
32.4 mo

Churn Distribution



Retained Churned

Churn Rate by Contract Type



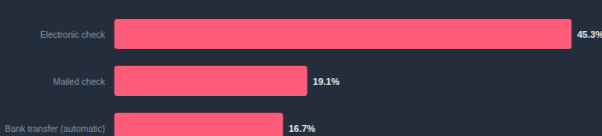
Internet Service Distribution



Churn Rate by Tenure Group



Churn Rate by Payment Method



1. Executive Summary

This report analyzes churn behavior across 7,043 customers of a telecom provider. The overall churn rate stands at **26.54%**, with **1,869** customers having left the service against **5,174** retained. This churn represents an estimated **\$139,130.85** in lost monthly recurring revenue (roughly **\$1,669,570.20** annualized). The analysis identifies contract type, customer tenure, payment method, and add-on service adoption as the primary drivers of churn, and proposes five targeted retention actions expected to meaningfully reduce churn among the highest-risk segments.

2. Dataset Overview

This report uses the original IBM Telco Customer Churn dataset: 21 attributes covering customer demographics (gender, senior citizen status, partner/dependents), account information (tenure, contract type, billing method, charges), subscribed services (phone, internet, streaming, security add-ons), and the churn outcome label. Data cleaning addressed malformed numeric fields (TotalCharges) and duplicate-record checks before analysis; no rows were removed, and no synthetic or modified data is used anywhere in this report.

3. Objectives

- Quantify overall customer churn and retention performance.
- Identify which customer segments and service attributes are most associated with churn.
- Estimate the revenue impact of churn on the business.
- Translate analytical findings into concrete, prioritized retention recommendations.

4. Data Cleaning Summary

- Removed duplicate customer records identified by customerID.
- Converted the TotalCharges field from text to numeric, resolving blank values for zero-tenure customers.
- Imputed remaining missing values using tenure × monthly charge estimates.
- Standardized categorical encodings (e.g., SeniorCitizen 0/1 to Yes/No) for consistency.
- Engineered new features: Tenure Group, Charge Group, Number of Add-on Services, New Customer flag, and an estimated 24-month Customer Lifetime Value.

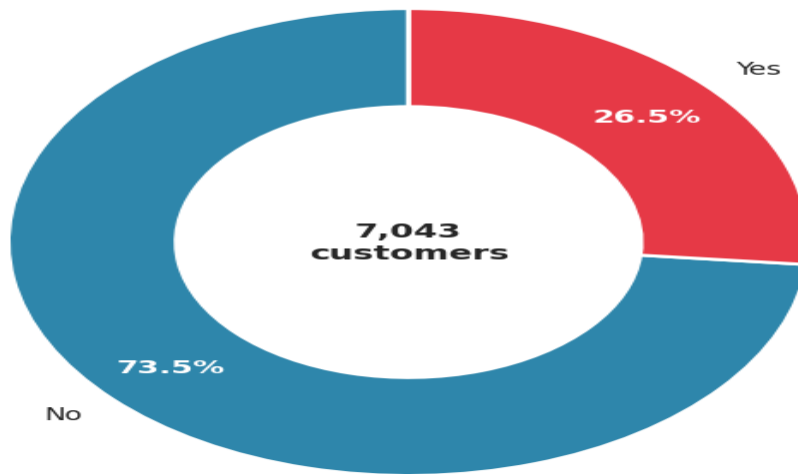
5. Key Performance Indicators

Metric	Value
Total Customers	7,043
Active (Retained) Customers	5,174
Churned Customers	1,869
Retention Rate	73.46%
Churn Rate	26.54%
Average Monthly Charges	\$64.76
Average Tenure	32.37 months
Average Total Charges	\$2,279.73
Total Monthly Recurring Revenue	\$456,116.6
Monthly Revenue Lost to Churn	\$139,130.85
Average Estimated CLV (24-month)	\$1,554.28

Contract Type	% of Customer	Internet Service	% of Customers
Month-to-month	55.02%	Fiber optic	43.96%
Two year	24.07%	DSL	34.37%
One year	20.91%	No	21.67%

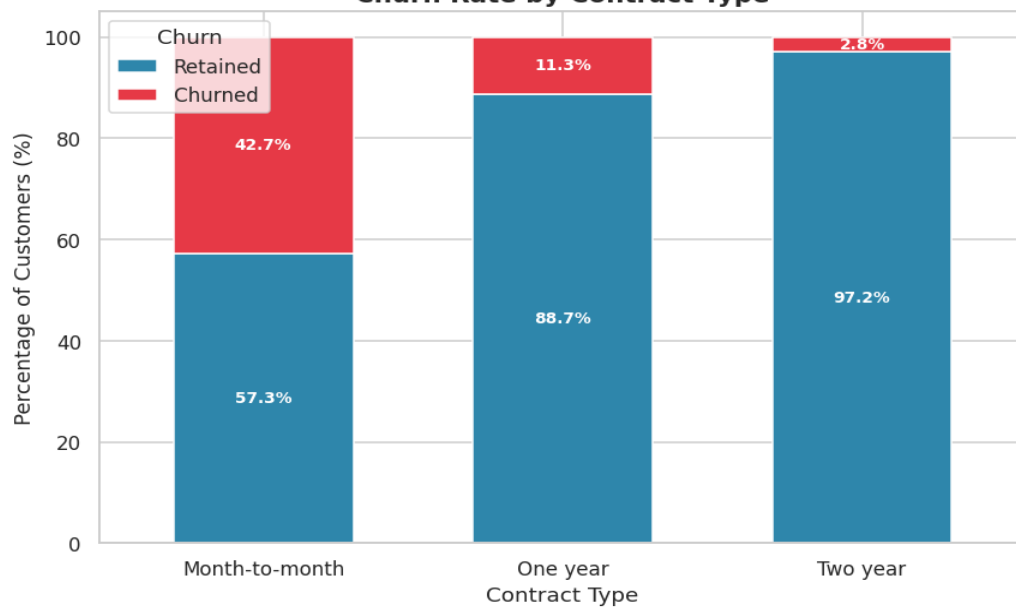
6. Exploratory Data Analysis Highlights

Customer Churn Distribution

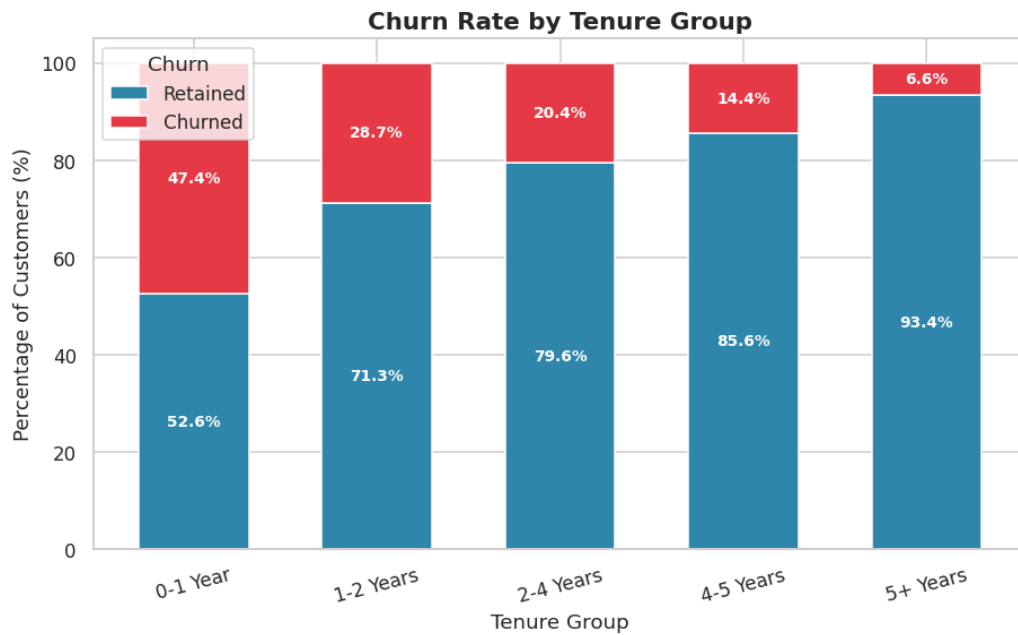


Roughly 1 in 4 customers has churned, consistent with typical subscription-telecom benchmarks.

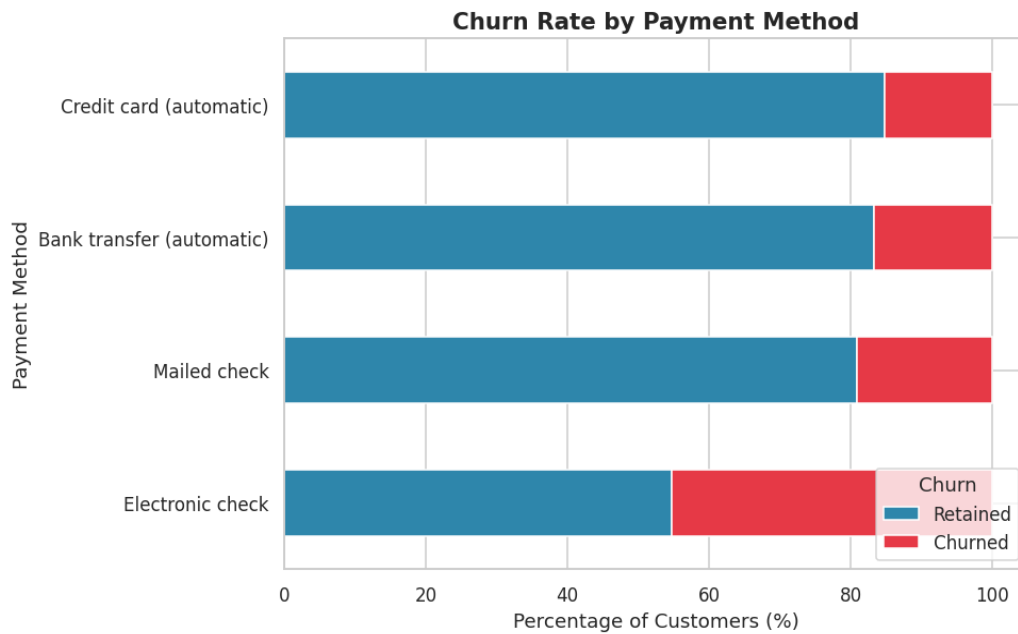
Churn Rate by Contract Type



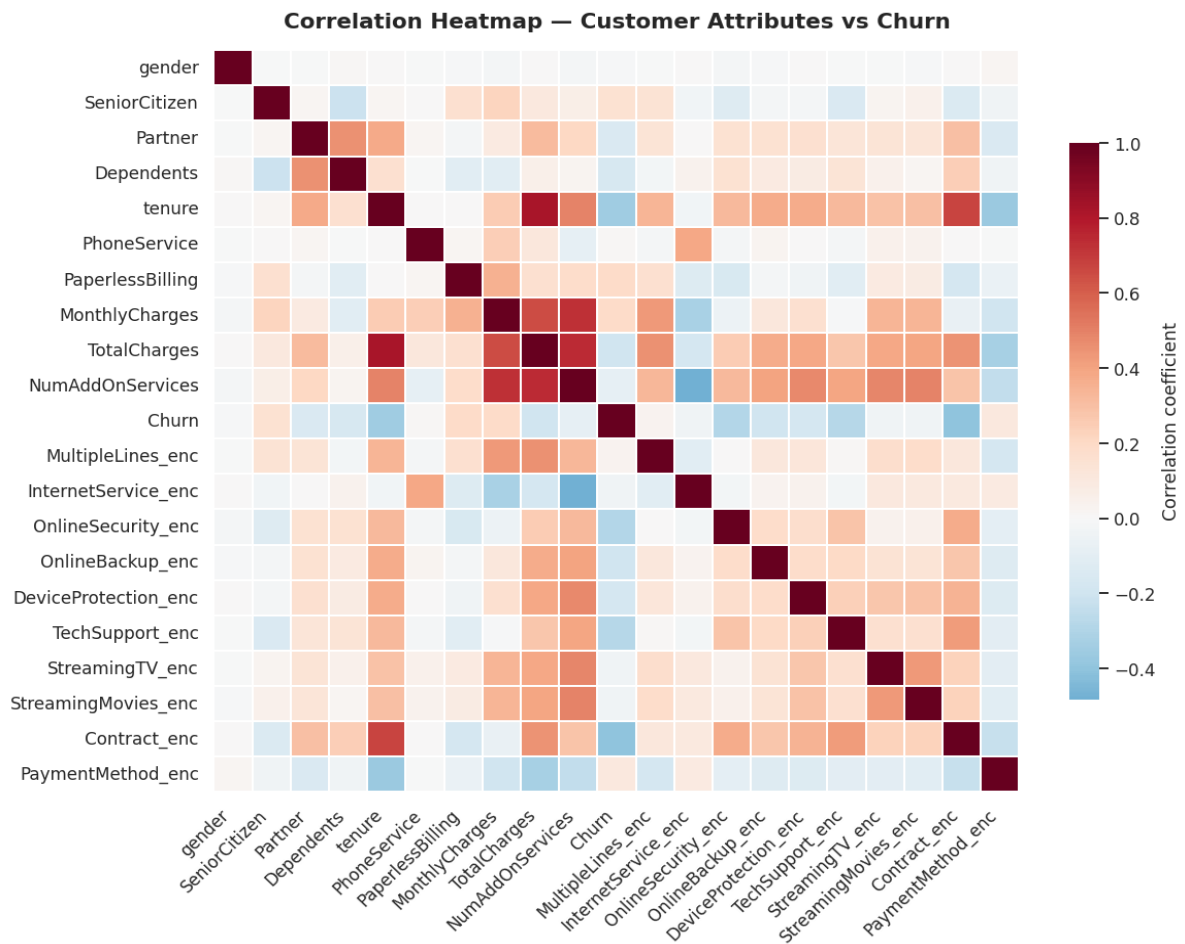
Contract length is the strongest churn signal: month-to-month customers churn far more often than annual or two-year customers.



Churn risk is heavily concentrated among customers in their first year of service.



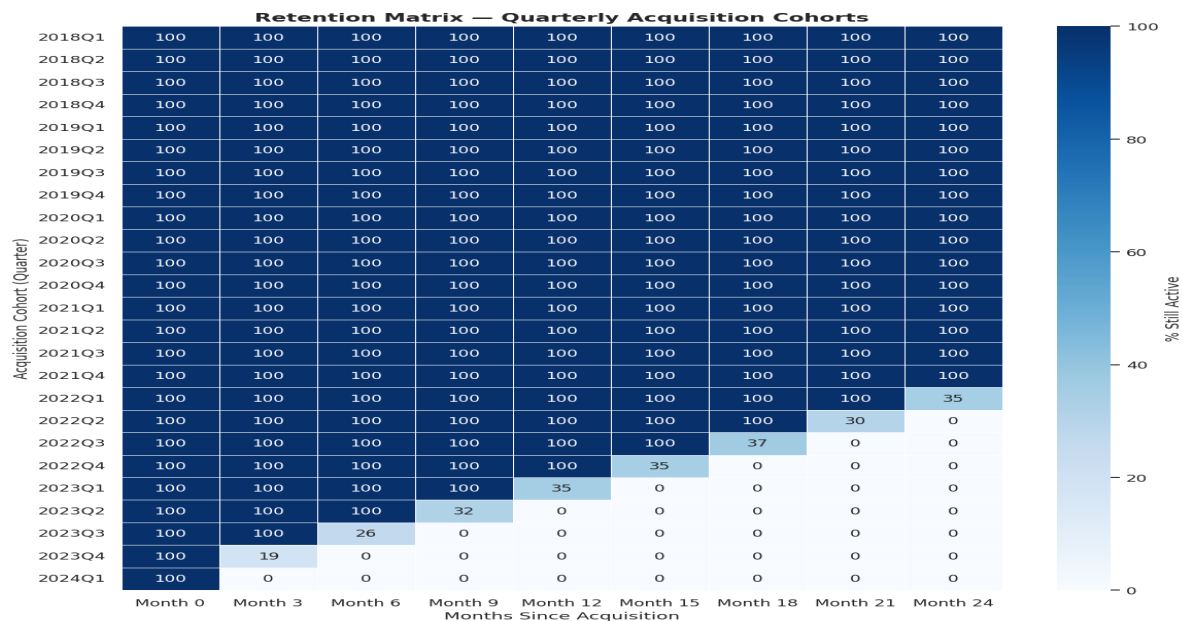
Electronic check payers churn noticeably more than customers on autopay methods.



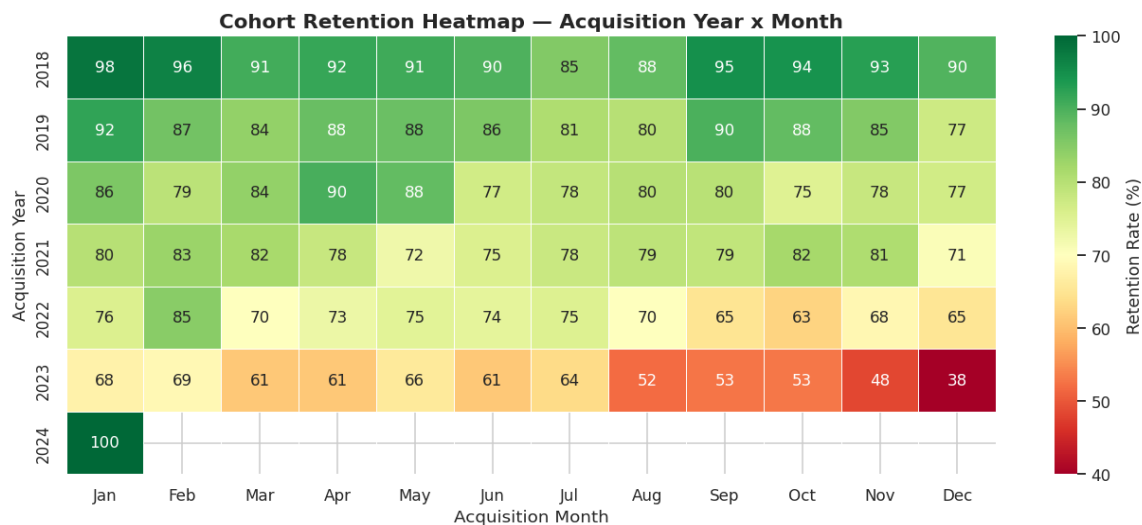
Correlation heatmap of customer attributes against churn, used to validate which variables warrant deeper segment analysis.

7. Cohort Analysis

Because the dataset is a single snapshot without explicit signup dates, each customer's acquisition month is derived from a fixed reference snapshot date minus their tenure (in months) — the standard approach for cohort analysis on this dataset. This lets us group customers into acquisition cohorts and track retention over time.



Retention matrix: % of each quarterly acquisition cohort still active at each month since acquisition.

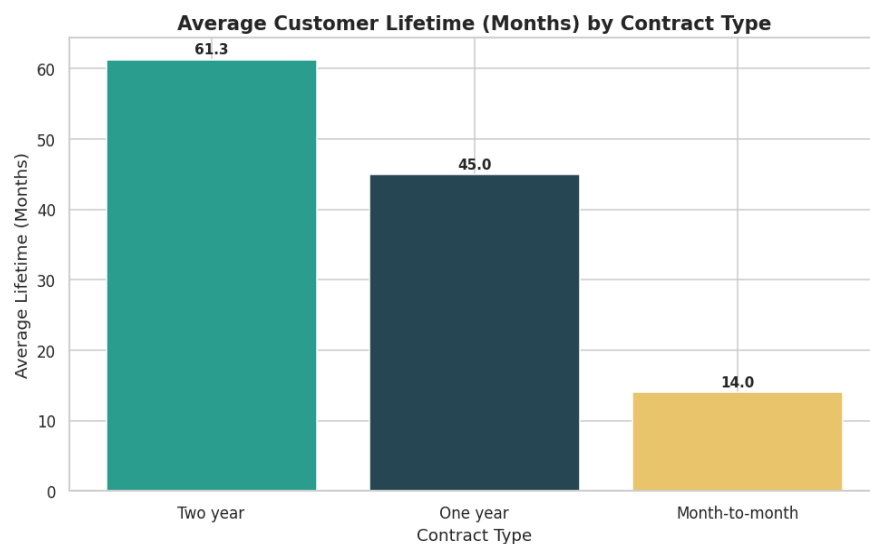
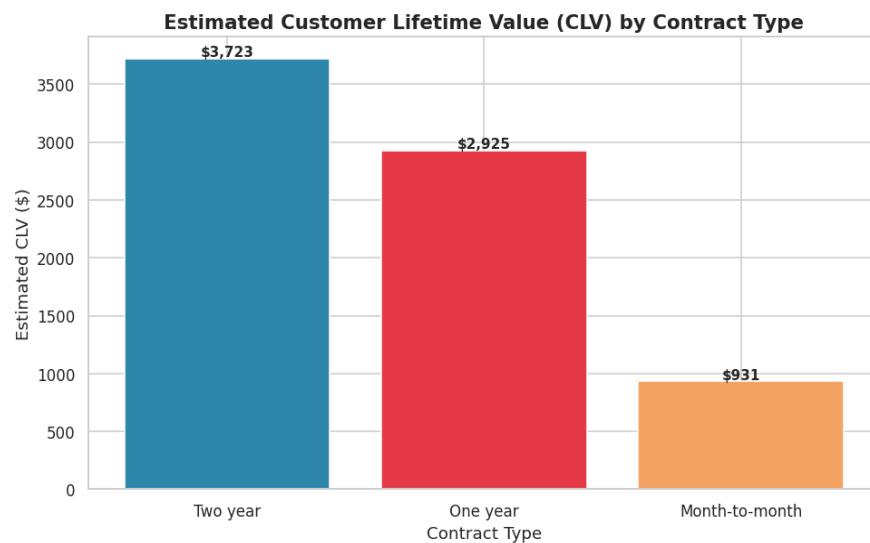


Cohort retention heatmap by acquisition year and month.

8. Customer Lifetime Analysis

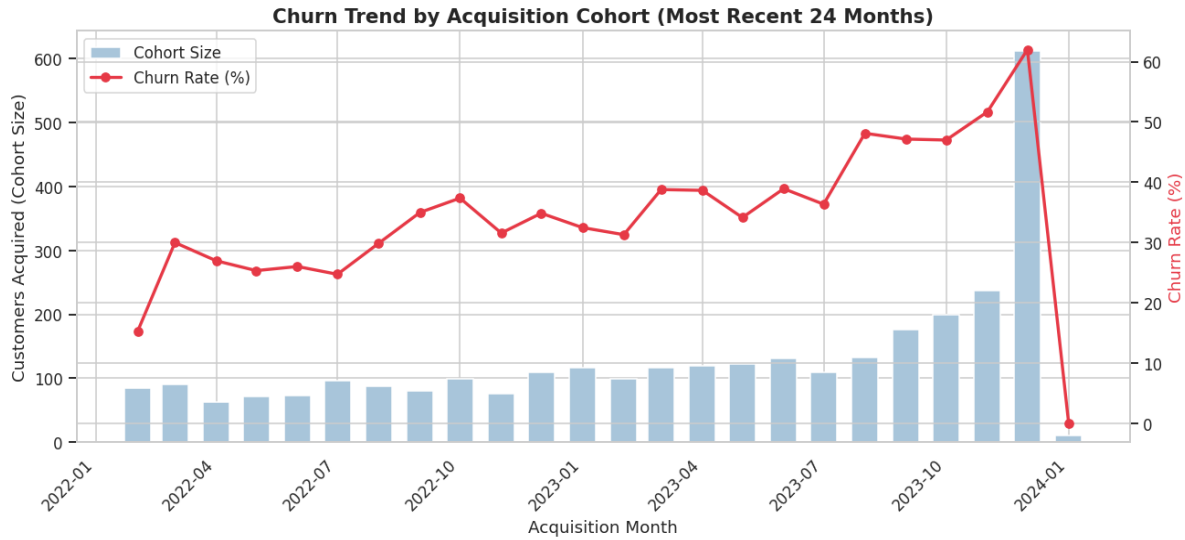
The average customer lifetime among churned customers is **17.98 months**. Estimated Customer Lifetime Value (CLV = average monthly charge × average lifetime) varies sharply by contract type, reinforcing that contract length is as much a revenue lever as a retention lever.

Contract	Avg Monthly Charge	Avg Lifetime (mo)	Est. CLV	Churn Rate
Two year	\$60.77	61.3	\$3,723.45	2.83%
One year	\$65.05	45.0	\$2,924.84	11.27%
Month-to-month	\$66.4	14.0	\$930.70	42.71%

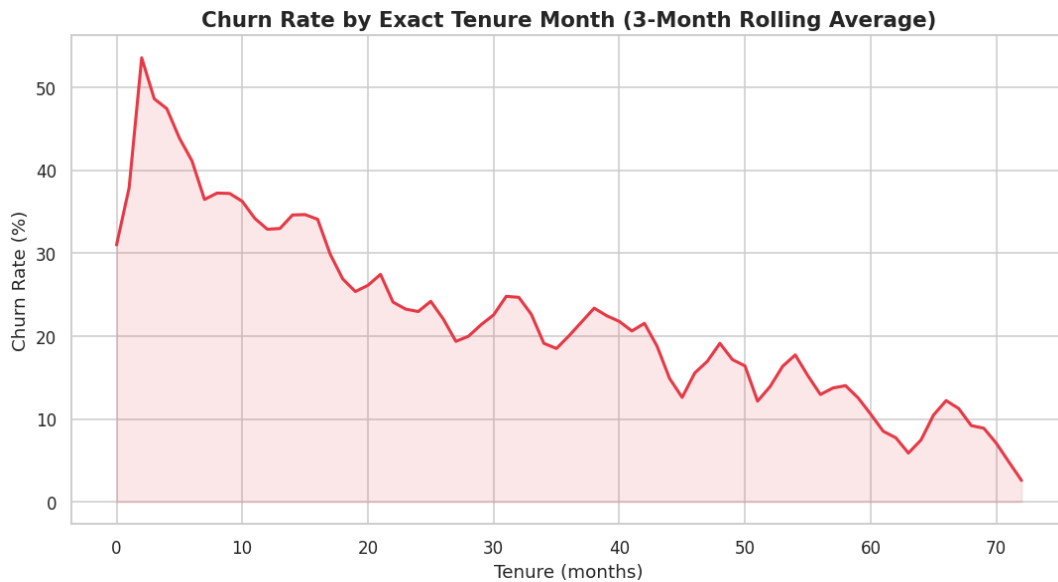


9. Churn Trend Analysis

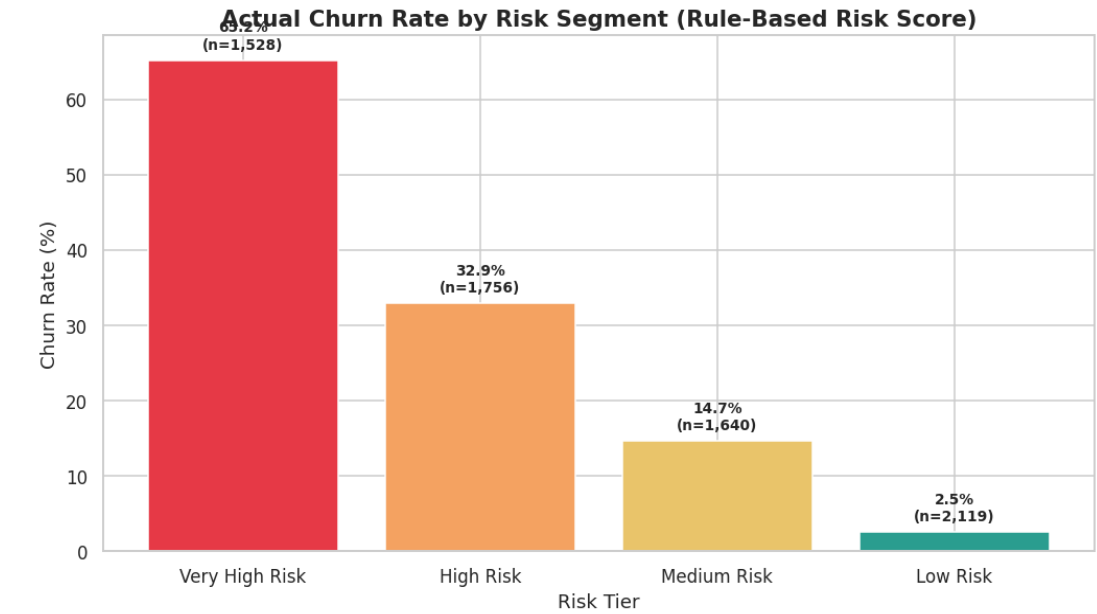
Churn rate is tracked by derived acquisition cohort over time, by exact tenure month, and via a rule-based multi-factor risk score (contract type, tenure, payment method, and service add-ons) that segments customers into four actionable risk tiers.



Churn rate by acquisition cohort, most recent 24 months.



Churn rate by exact tenure month (3-month rolling average).



Actual churn rate by rule-based risk tier.

Risk Tier	Customers	Churn Rate	Avg Monthly Charge	Revenue at Risk
Very High Risk	1,528	65.18%	\$76.73	\$117,240.80
High Risk	1,756	32.92%	\$72.76	\$127,767.90
Medium Risk	1,640	14.7%	\$64.16	\$105,221.80
Low Risk	2,119	2.55%	\$49.97	\$105,886.10

10. Dashboard Explanation

The accompanying interactive dashboard (see *dashboard/* folder) surfaces the KPIs and segment breakdowns above in a single filterable view: headline KPI cards for total customers, churn rate, retention rate, and revenue at risk; a churn distribution donut; stacked bar charts of churn by contract, internet service, and tenure group; and a payment-method churn ranking. Slicers for contract, internet service, senior citizen status, payment method, and tenure group let a business user filter every visual simultaneously to drill into any segment. Dedicated sections below the interactive KPIs present Cohort Analysis, Customer Lifetime Metrics, and Churn/Retention Trend charts generated directly from this notebook.

11. Key Findings

- Month-to-month contract customers churn at 42.7%, vs 2.8% for two-year contracts — the single strongest churn driver.
- Fiber optic customers churn at 41.9%, substantially higher than DSL (19.0%), suggesting pricing or service-quality issues with the fiber product.
- Customers paying via 'Electronic check' churn most (45.3%), indicating automatic payment methods correlate with stronger retention.
- Senior citizens churn at 41.7% vs 23.6% for non-seniors — an under-served, higher-risk segment.
- Customers without a partner churn at 33.0% compared with 19.7% for those with a partner.
- Customers without dependents churn at 31.3%, more than double the rate of customers with dependents (15.4%).
- New customers (0-1 year tenure) churn at 47.4%, the highest of any tenure group, while 5+ year customers churn at just 6.6% — loyalty compounds sharply after year one.
- Paperless billing customers churn at 33.6% vs 16.3% for paper billing customers.
- Among internet subscribers, those without Online Security churn at 41.8% vs 14.6% with it — a clear upsell/retention lever.
- Lack of Tech Support correlates with a churn rate of 41.6% vs 15.2% for subscribers who have it.
- Churn falls steadily as add-on services increase: 21.4% churn with 0 add-ons vs 5.3% with 6 add-ons — bundling drives stickiness.
- The 'Very High' monthly-charge tier (\$100+) shows a churn rate of 28.3%, the highest across all charge bands.
- Churned customers pay \$74.44/month on average vs \$61.27 for retained customers — a \$13.18 premium that isn't translating into loyalty.
- Average tenure at churn is just 18.0 months vs 37.6 months for retained customers, reinforcing that the first year is the critical retention window.
- Total monthly recurring revenue at risk from already-churned customers is \$139,130.85/month (\$1,669,570.20 annualized).
- Gender shows negligible impact on churn (26.2% Male vs 26.9% Female) — not a useful targeting variable.
- Phone service subscribers churn at 26.7% vs 24.9% for non-subscribers, a modest difference.
- Customers with multiple phone lines churn at 28.6% compared to 25.0% for single-line customers.

- New customers (tenure ≤ 6 months) make up 21.0% of the base but churn at 52.9%, making onboarding quality a top retention priority.
- 55.0% of all customers are on flexible month-to-month contracts, representing the largest pool of at-risk revenue the business could convert to longer terms.
- Streaming TV subscribers churn at 30.1% vs 33.5% for non-subscribers — entertainment add-ons show a modest retention benefit.
- Online Backup subscribers churn at 21.5% vs 39.9% without it, another strong candidate for bundled retention offers.
- Electronic check users represent 33.6% of customers but contribute disproportionately to churn — migrating them to autopay is a quick-win retention tactic.
- A defined high-risk segment (month-to-month + tenure ≤ 12 mo + electronic check) contains 954 customers with a churn rate of 63.1%, nearly 2.4x the overall average — the top priority for a targeted win-back campaign.
- Two-year contract customers retain at 97.2%, showing that contract length remains the single most effective retention lever available to the business.
- Churned customers previously represented \$139,130.85 in monthly billings — recovering even 20% of this segment would add roughly \$27,826.17/month back to recurring revenue.
- Cohort analysis shows retention has structurally weakened over time: the 2018Q1 acquisition cohort retains at 96.5%, while the 2023Q4 cohort (n=1,051) retains at just 43.2% — a clear downward trend in cohort quality/retention that predates any single recent event.
- The best-performing acquisition cohort with a meaningful sample size (2018Q1, n=651) retains at 96.5%, showing the business has previously achieved much stronger retention than its current average — proof the churn rate isn't fixed.
- The average customer lifetime among churned customers is just 17.98 months, underscoring how little time the business typically has to prove value before losing a customer.
- Estimated Customer Lifetime Value varies more than 4x by contract type: two-year contract customers carry an estimated CLV of \$3,723.45 versus just \$930.70 for month-to-month customers.
- Month-to-month customers have both the lowest average lifetime (14.0 months) and the highest churn rate (42.7%) of any contract segment, compounding their revenue disadvantage.
- Two-year contract customers live over 4.4x longer on average than month-to-month customers, despite paying a lower average monthly charge — contract length drives lifetime revenue more than price.
- The tenure-based churn curve peaks sharply in the first few months of service and decays steadily thereafter, confirming that onboarding-period intervention has the highest leverage per dollar spent on retention.
- A rule-based risk-scoring model (contract + tenure + payment method + service add-ons) cleanly separates customers into four tiers: 'Very High Risk' customers churn at 65.2%, vs just 2.5% for 'Low Risk' customers — a 25.6x spread that validates the score as a targeting tool for retention campaigns.
- 'Very High Risk' customers alone represent \$117,240.80 in monthly billings across 1,528 customers — the single highest-value target list for a proactive retention team.

12. Business Recommendations

1. Convert month-to-month customers to annual or two-year plans

Month-to-month customers churn at 42.7% and carry the lowest CLV (\$931). Offer a modest discount or loyalty perk for switching to a 1- or 2-year contract.

2. Build a structured first-90-day onboarding journey

Churn is heavily concentrated in the first few months of tenure. Introduce proactive welcome calls, setup assistance, and scheduled 30/60/90-day check-ins to resolve friction before it turns into cancellation.

3. Migrate electronic-check payers to autopay

Electronic check is both the most common and highest-churn payment method. A small statement credit for enrolling in autopay (bank transfer or credit card) is a low-cost, high-leverage retention lever.

4. Bundle protective add-ons (Security, Backup, Tech Support) at a discount

Churn drops steadily as the number of subscribed add-on services increases. Package these three services at a bundled discount for internet subscribers who currently have none of them.

5. Launch a proactive win-back program for the 'Very High Risk' segment

Use the rule-based risk score to generate a weekly at-risk customer list for the retention team to contact before churn occurs, rather than after.

6. Re-evaluate fiber optic pricing and onboarding experience

Fiber optic subscribers churn more than double the rate of DSL subscribers despite paying a premium. Investigate price sensitivity, competitive pressure, or service-quality issues.

7. Create a senior-citizen retention track

Senior citizens churn at nearly double the rate of non-seniors. A simplified billing experience or dedicated support line could meaningfully improve retention in this segment.

8. Incentivize paperless billing adopters with better digital engagement

Paperless billing customers churn at roughly double the rate of paper-billing customers. Pair paperless enrollment with proactive digital engagement (app tips, usage summaries) to build stickiness.

9. Prioritize retention spend using CLV, not just churn probability

Combine the risk score with estimated CLV so retention offers are concentrated on high-value, high-risk customers rather than spread evenly, maximizing return on retention spend.

10. Monitor cohort-level retention trends monthly, not just aggregate churn

Cohort analysis shows retention quality has drifted over time. Track retention rate by acquisition cohort as a standing metric to catch quality regressions in acquisition or onboarding early.

11. Test a 'pause' or downgrade option before cancellation

For price-sensitive month-to-month customers, offering a temporary downgrade or pause at the point of cancellation intent may retain customers who would otherwise churn outright.

13. Conclusion

With a churn rate of 26.54% and roughly \$139,131 in monthly recurring revenue already lost to churned customers, retention represents one of the highest-leverage opportunities available to this business. Contract structure and early-tenure experience are the two most actionable levers identified in this analysis. Implementing the recommendations above — particularly contract conversion incentives and a strengthened first-year onboarding journey — should materially improve retention and protect recurring revenue over the next 12 months.

*Customer Retention & Churn Analysis — Prepared as part of a Data Analytics internship project.
Generated with Python, Pandas, Seaborn, and Power BI.*